Technical Explanation

Structure-aware document chunking for the Aurelia Digital Finance Ltd. integrated financial and sustainability report.

Objective and outcome

The objective was to convert the supplied nine-page PDF into retrieval-ready records while preserving the relationships among headings, prose, lists, tables, financial values, and long policy language. The final run processed 9 pages and generated 29 JSONL chunks, including 8 table chunks, 2 bullet-list chunks, and 3 policy-clause chunks. Ten distinct records were selected from the generated output as representative examples.

Architecture and workflow

The command-line entry point calls four focused modules: pdf_extractor reads pages and tables; structure_parser detects numbered and named headings and classifies content; chunk_builder creates bounded chunks and overlap; validators checks records and reparses JSONL. Input and output locations are supplied through argparse, so no absolute machine path is embedded in the project.

Libraries and extraction strategy

pypdf validates page-wise access and pdfplumber supplies coordinate-aware lines and tables. Events are sorted by vertical position, so each table receives the heading context that precedes its own bounding box rather than the last heading on its page. Characters inside a detected table box are excluded from prose extraction, reducing duplicates. The parser removes only the repeated report banner and page label identified as boilerplate.

Chunking and token decisions

The requested target was 400 tokens with a 60-token preferred overlap and a 550-token maximum. This real report produced 29 useful chunks at the requested target, so no target reduction was required. Structural boundaries kept all stored chunks below the target, so the configured overlap was not inserted in this run. Token counts use tiktoken cl100k_base. The builder adds a sentence-tail overlap only when splitting requires it, and uses a word-boundary fallback only when one sentence exceeds the maximum.

Structure-specific handling

Numbered, named, and appendix headings are normalized safely and stored in section metadata. The Key Highlights and Board Meeting Disclosure artifacts are normalized into bullet-list content. Tables are independent Markdown chunks with their header row and values together; the two page-3 tables map to Consolidated Results and Segment Performance respectively, and compliance metrics remain outside the customer-data clause. Long clauses are classified as policy_clause and split only at sentence boundaries when necessary. Overlap is recorded only when the exact sentence tail was injected; independent table chunks report zero.

Metadata and retrieval benefits

Every record includes deterministic chunk_id, document_id, document_name, chunk_index, page_number, page_numbers, section, section_path, content_type, text, token_count, token_count_method, and overlap_tokens. These fields support page-cited retrieval, filtering by section or content type, deterministic reprocessing, and evaluation of whether a retrieved answer came from a table, policy clause, or narrative.

Representative observations

The output covers company overview, financial performance, risk management, governance, sustainability and ESG, regulatory compliance, management outlook, both appendices, and the final fictional-dataset disclosure. It

preserves INR 1,842.6 crore revenue, the risk table, renewable-electricity metrics, and compliance metrics. The summary records no extraction warnings and confirms that all 29 JSONL records were reparsed successfully.

Validation performed

The complete command-line pipeline was executed against the supplied PDF. JSONL was reparsed after writing and every record was checked for required fields, non-empty text, unique IDs, valid physical page numbers, and honest overlap metadata. The pytest suite was executed with 13 passing tests. Tests cover policy clauses flushed before headings or tables, table-to-section associations, named headings, bullet normalization, independent-table overlap, full-output coverage, preserved financial and ESG values, and stable IDs over repeated runs.

Limitations and production improvements

PDF structure remains heuristic. Image-only documents, irregular multi-column layouts, malformed text encodings, merged cells, and tables spanning pages can require OCR, layout-model support, or review rules. A production RAG system could add OCR fallback, coordinate-level provenance, cross-page table stitching, embedding and retrieval evaluation, semantic reranking, configurable document profiles, and sampled human quality review. These additions should be evaluated against the document corpus rather than assumed to be universally beneficial.

Executed configuration	Result
Pages processed	9
Chunks generated	29
Representative chunks	10
Target / overlap / maximum	400 / 60 / 550 tokens
Actual overlap	0 tokens; structural boundaries used
Token method	tiktoken cl100k_base
Validation	JSONL valid; 13 pytest tests passed